

Module 12: Darwin and Evolution - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Natural selection requires all of the following EXCEPT:

- A. variation among individuals
- B. traits that are heritable
- C. every individual reproducing at an equal rate
- D. differences in survival or reproduction
- Answer: C
- Why: Selection needs variation, heritability, and differences in reproductive success. Equal reproduction by all individuals is exactly what selection is not.

2. In evolutionary terms, an organism's "fitness" means its:

- A. relative reproductive success in its environment
- B. physical strength
- C. overall body size
- D. top running speed
- Answer: A
- Why: Fitness is about passing on genes through reproductive success, not about athletic ability.

3. The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:

- A. identical environments
- B. no relationship at all
- C. learning during each animal's life
- D. common ancestry shown by homologous structures
- Answer: D
- Why: Homologous structures share an inherited body plan from a common ancestor, which supports common descent.

4. "An individual giraffe stretched its neck during its life and so it evolved a longer neck."

Why is this wrong?

- A. Giraffes actually have short necks.
- B. Individuals do not evolve; populations evolve as heritable trait frequencies change over generations.
- C. Evolution is completed within a single lifetime.
- D. Stretching directly rewrites an animal's DNA.
- Answer: B
- Why: Evolution is a population-level change across generations, not a change acquired within one individual's lifetime.